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Studierichting: Informatica
Oefeningenreeks 3F nr. 2.5

Bepaal de schuine asymptoot door een euclidische deling te maken:

$$f(x) = \frac{\sqrt{2}x^2 + \sqrt{3}x - \sqrt{2}}{x + \sqrt{6}}$$

$$\begin{array}{cccc|l}
 & +\sqrt{2}x^2 & +\sqrt{3}x & -\sqrt{2} & x + \sqrt{6} \\ \cline{5-5}
- & +\sqrt{2}x^2 & +2\sqrt{3}x & & \sqrt{2}x - \sqrt{3} \\ \cline{2-3}
 & & -\sqrt{3}x & -\sqrt{2} & \\
- & & -\sqrt{3}x & -3\sqrt{2} & \\ \cline{3-4}
 & & & +2\sqrt{2} &
\end{array}$$

Schuine asymptoot:

$$A : y = \sqrt{2}x - \sqrt{3}$$

$$f(x) - A = \frac{2\sqrt{2}}{x + \sqrt{6}}$$

$$\begin{array}{c|ccc}
x & & -\sqrt{6} & \\ \hline
f(x) - A & - & | & +
\end{array}$$

Dus:

$$\lim_{x \to +\infty} (f(x) - A) = 0^{+} \Rightarrow x \to +\infty : f > A$$

$$\lim_{x \to -\infty} (f(x) - A) = 0^{-} \Rightarrow x \to -\infty : f < A$$

References